$$N_k = \sum_{i+j=k} \binom{15}{i} 3^{\,i} \binom{2}{j} 2^{\,j} 2^{\,2-j} \qquad (0 \le i \le 15,\ 0 \le j \le 2)$$

$$p = \frac{1}{4^{17}} \sum_{k=0}^{5} N_k = \frac{4\,458\,112}{4^{17}} \approx 2.6 \times 10^{-4} \approx \frac{1}{3854}$$

$$E = D \cdot p$$